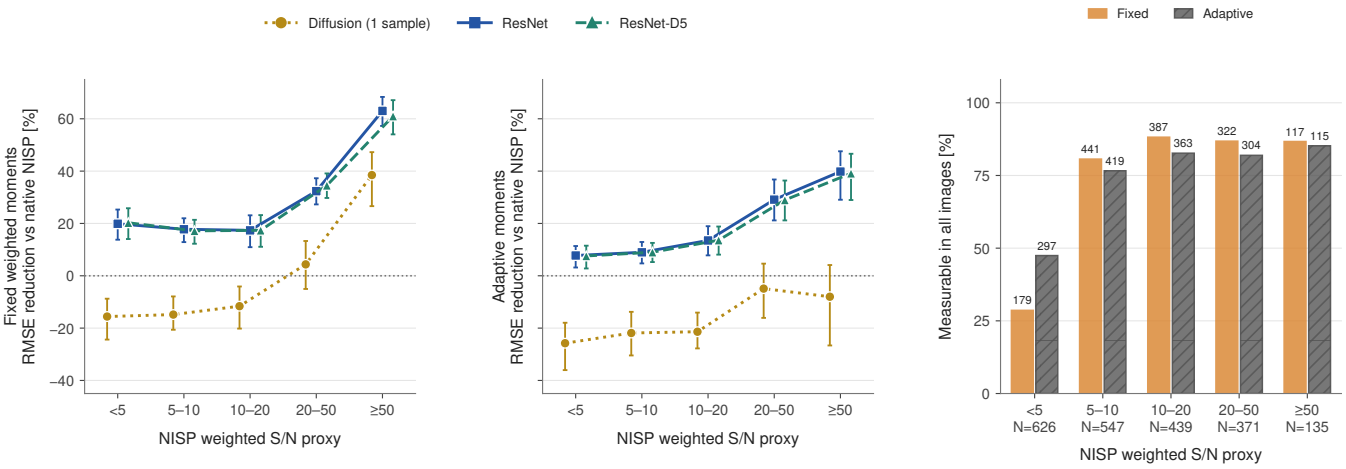


Figure 8: Observed-shape recovery versus input NISP weighted S/N. Left and middle: percentage reduction in ellipticity-component RMSE relative to native NISP, for fixed weighted and adaptive moments, respectively; positive values indicate lower error against observed JWST F115W. Ochre circles, cobalt squares, and green triangles denote a single diffusion realization, ResNet, and ResNet-D5. Each estimator/bin uses the same galaxies measurable in JWST, native NISP, and all six reconstruction/interpolation methods. Points are horizontally offset for visibility; the bins are categorical and connecting lines guide the eye. Whiskers show conditional 95% intervals from 2000 paired spatial-block resamples, not simultaneous intervals for a trend. Right: common measurement coverage among all attempted fields, with fixed moments in solid orange and adaptive moments in hatched charcoal. Numbers above bars are common successes; N below each bin gives the attempted count. Totals are 1446 fixed-moment and 1498 adaptive-moment successes out of 2118 validation fields. The S/N proxy neglects correlated noise, and fixed and adaptive moments have different successful samples. These observed cross-band comparisons include PSF and pixel-sampling differences and do not measure calibrated shear.

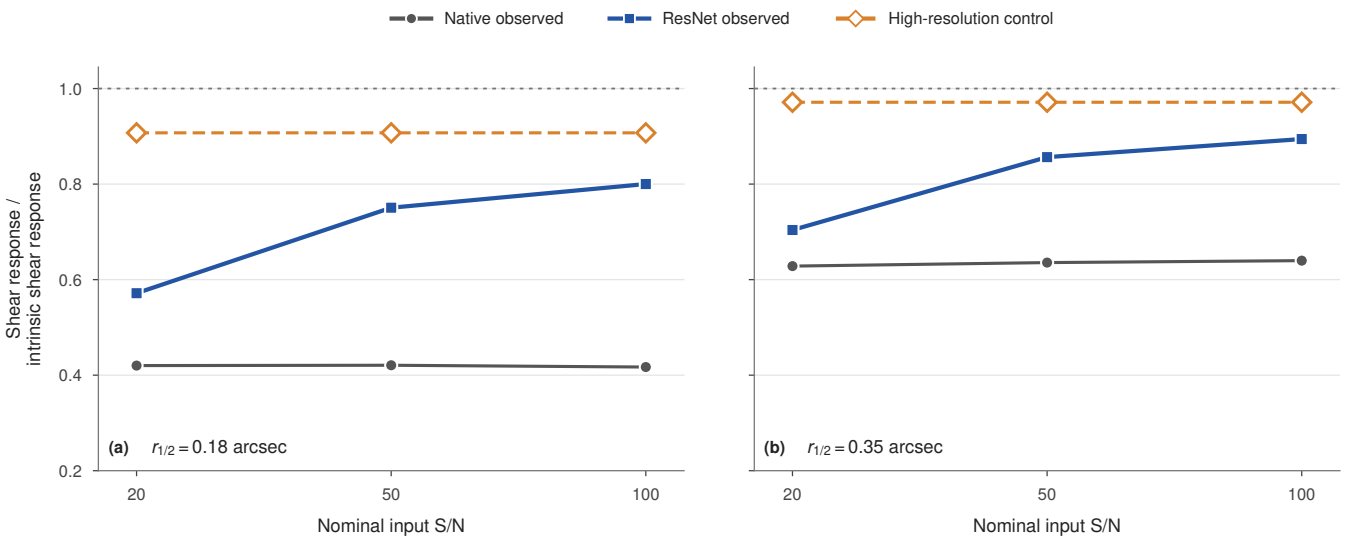


Figure 9: Controlled shear response of observed adaptive shapes using the four supplied NISP effective PSFs. Panels (a) and (b) show exponential galaxies with intrinsic half-light radii 0.″18 and 0.″35. Charcoal circles, cobalt squares, and orange open diamonds denote native NISP, the original ResNet, and a noiseless high-resolution control. The vertical axis is $\mathrm{tr}(R)/\mathrm{tr}(R_{\mathrm{intrinsic}})$; the dotted line marks the intrinsic response. Each point equally averages the four per-PSF ratios on identical objects successful across all PSFs, methods, and five shear branches. Each noisy cell attempts eight orientations and four noise realizations per orientation; the common sample is 31/32 at radius 0.″18 and S/N 20, and 32/32 elsewhere in the displayed grid. The same noise is reused across shear branches and PSFs. S/N positions are categorical and connecting lines guide the eye; these are point estimates without uncertainty intervals. The high-resolution control uses a Gaussian PSF of FWHM 0.″08, not a validated JWST PSF. The two radii were selected for their larger gains in the earlier Gaussian-PSF pilot; the full new grid also includes radius 0.″65 and noiseless native inputs. No PSF correction is applied to these shapes, and the responses do not establish calibrated weak-lensing accuracy.